13/4/2024 Kuhn-Tucker-Conditions

(1) For solving NLP Constraint = h

2 variable and one constraint

$$\frac{\partial L}{\partial x_1} = 0 \longrightarrow (1)$$

$$\frac{\partial L}{\partial x_2} = 0 \longrightarrow (2)$$

$$\lambda(h) = 0 \longrightarrow (3)$$

$$h \leq 0 \longrightarrow (4)$$

$$x_1, x_2 \geq 0 \longrightarrow (5)$$

$\lambda \geq 0$ max

$\lambda \leq 0$ Min

eq (1)(2) & (3) used For calculation and (4) and (5) and

2 variables and 2 constraints

$$\frac{\partial L}{\partial x_1} = 0 \longrightarrow (1)$$

$$\frac{\partial L}{\partial x_2} = 0 \longrightarrow (2)$$

$$\lambda_1(h_1) = 0 \; (3) \quad \lambda_2(h_2) = 0 \longrightarrow (4)$$

$$h_1 \leq 0 \longrightarrow (5) \quad h_2 \leq 0 \longrightarrow (6)$$

$$x_1, x_2 \geq 0 \longrightarrow (7)$$

$\lambda_1, \lambda_2 \geq 0$ — max

$\lambda_1, \lambda_2 \leq 0$ — min

2

Kuhn-Tucker Conditions

NLPP

Prob. Max $z = 2x_1^2 - 7x_2^2 + 12x_1x_2$

Sub to $2x_1 + 5x_2 \leq 98$

Kuhn-Tucker Conditions

$\frac{\partial L}{\partial x_1} = 0$ — (1)

$\frac{\partial L}{\partial x_2} = 0$ — (2)

$\lambda(h) = 0$ — (3)

$h \leq 0$ — (4)

$x_1, x_2 \geq 0$ — (5)

$\lambda \geq 0$ — max

Step 01

$L(x_1, x_2, \lambda) = (2x_1^2 - 7x_2^2 + 12x_1x_2) - \lambda(2x_1 + 5x_2 - 98)$

$h = 2x_1 + 5x_2 - 98$

$L = (2x_1^2 - 7x_2^2 + 12x_1x_2) - \lambda(2x_1 + 5x_2 - 98)$

$\frac{\partial L}{\partial x_1} = 4x_1 + 12x_2 - 2\lambda = 0$ — (1)

$\frac{\partial L}{\partial x_2} = -14x_2 + 12x_1 - 5\lambda = 0$ — (2)

$\lambda(2x_1 + 5x_2 - 98) = 0$ — (3)

$2x_1 + 5x_2 - 98 \leq 0$ — (4)

$x_1, x_2 \geq 0, \lambda \geq 0$ max — (5)

$4x_1 + 12x_2 - 2\lambda = 0$ — (1)

$-14x_1 - 12x_1 + 5\lambda = 0$ — (2)

eq (1) × 5 and eq (2) × 2 and sub.

$20x_1 + 60x_2 - \cancel{10\lambda} = 0$ — (6)

$\underline{\mp 24x_1 \mp 28x_2 \cancel{\mp 10\lambda} = 0}$ — (7)

$-4x_1 + 88x_2 = 0$ — (8)

$-4(x_1 - 22x_2) = 0$

$x_1 - 22x_2 = 0$ — (9) × 2 and sub from (3)

$2\cancel{x_1} + 5x_2 - 98 = 0$

$\underline{\mp 2\cancel{x_1} \pm 44x_2 + 0 = 0}$

$49x_2 - 98 = 0$

$49x_2 = 98$

$x_2 = \frac{98}{49}$

$\boxed{x_2 = 2}$ Put in (9)

$x_1 - 22(2) = 0$

$x_1 - 44 = 0$

$\boxed{x_1 = 44}$ Put in (1)

$4(44) + 12(2) - 2\lambda = 0$

$176 + 24 - 2\lambda = 0$

$2\lambda = 200$

$\lambda = \frac{200}{2}$

$\boxed{\lambda = 100}$

$\frac{01}{176}$ $\frac{24}{280}$

$\frac{49}{2}$ 98

CamScanner

(2)

Kuhn-Tucker Conditions

NLPP

Prob. Max $Z = 2x_1^2 - 7x_2^2 + 12x_1x_2$

sub to $2x_1 + 5x_2 \leq 98$

$$\begin{array}{r} 3872 \\ 1056 \\ \hline 4928 \\ 28 \\ \hline 4900 \end{array}$$

(3)

$x_1 = 44, \; x_2 = 2, \; \lambda = 100$

$Z = 2x_1^2 - 7x_2^2 + 12x_1x_2$

$Z = 2(44)^2 - 7(2)^2 + 12(44)(2)$

$Z = 2(1936) - 7(4) + 1056$

$Z = 3872 - 28 + 1056$

$\boxed{Z = 4900}$

Check's

$h \leq 0$

$2x_1 + 5x_2 - 98 \leq 0$

$2(44) + 5(2) - 98 \leq 0$

$88 + 10 - 98 \leq 0$

$98 - 98 \leq 0$

$0 \leq 0$

$x_1, x_2 \geq 0$

$44 \geq 0$

$2 \geq 0$

$\lambda \geq 0$ max

$100 \geq 0$ max

$$\begin{array}{r} 44 \\ \times 44 \\ \hline 176 \\ 176\times \\ \hline 1936 \\ \times 2 \\ \hline 3872 \end{array} \qquad \begin{array}{r} 44 \\ \times 2 \\ \hline 88 \\ \times 12 \\ \hline 1056 \end{array}$$

Exam Problem 2021 (5)

min Cost

$C = (x_1+4)^2 + (x_2-4)^2$

$x_1^2 + 16 - 2x_1$ $\quad 2(x_1)(4)$

$2x_1 - 2$ $\quad 8x_1$

$2(x_1-1)$

$x_1^2 + 16 - 8x_1$

$2x_1 - 8$

$2(x_1-4)$

$St \cdot 2x_1 + 3x_2 \geq 6$

$-3x_1 - 2x_2 \geq -12$

$\frac{\partial L}{\partial x_1} = 0$ — (1)

$\frac{\partial L}{\partial x_2} = 0$ — (2)

$\lambda_1(h_1) = 0$ — (3)

$\lambda_2(h_2) = 0$ — (4)

$h_1 \leq 0$ — (5)

$h_2 \leq 0$ — 6

$x_1, x_2 \geq 0$ — (7)

$\lambda_1 \lambda_2 \leq 0$ min

$L(x_1, x_2, \lambda_1, \lambda_2)$

$L = (x_1-4)^2 + (x_2-4)^2 - \lambda_1(2x_1 + 3x_2 - 6) - \lambda_2(-3x_1 - 2x_2 + 12)$

$h_1 = 2x_1 + 3x_2 - 6$

$h_2 = -3x_1 - 2x_2 + 12$

$\frac{\partial L}{\partial x_1} = 2(x_1-4) - 2\lambda_1 + 3\lambda_2 = 0$ — (1)

$\frac{\partial L}{\partial x_2} = 2(x_2-4) - 3\lambda_1 + 2\lambda_2 = 0$ — (2)

$\lambda_1(2x_1 + 3x_2 - 6) = 0$ — (3)

$\lambda_2(-3x_1 - 2x_2 + 12) = 0$ — (4)

$2x_1 + 3x_2 - 6 \leq 0$ — (5)

$-3x_1 - 2x_2 + 12 \leq 0$ — (6)

$x_1, x_2 \geq 0$

$\lambda_1, \lambda_2 \leq 0$ min

CamScanner

Problem - 3 **Kuhn-Tucker conditions** 6

$$\max \quad Z = 10x_1 + 10x_2 - x_1^2 - x_2^2$$

$$\text{sub to: } x_1 + x_2 \leq 8$$

$$-x_1 + x_2 \leq 5$$

$\frac{\partial L}{\partial x_1} = 0$ — (1)

$\frac{\partial L}{\partial x_2} = 0$ — (2)

$\lambda_1 (h_1) = 0$ — (3)

$\lambda_2 (h_2) = 0$ — (4)

$h_1 \leq 0$ — (5)

$h_2 \leq 0$ — (6)

$x_1, x_2 \geq 0$ — (7)

$\lambda_1, \lambda_2 \geq 0$ max — (8)

From (3) and (4)

$$\begin{aligned} x_1 + x_2 - 8 &= 0 \\ -x_1 + x_2 - 5 &= 0 \end{aligned}$$

$\cancel{2x_2 - 13 = 0}$

$L(x_1, x_2, \lambda_1, \lambda_2)$

$$L = (10x_1 + 10x_2 - x_1^2 - x_2^2) - \lambda_1 (x_1 + x_2 - 8) - \lambda_2 (-x_1 + x_2 - 5)$$

$h_1 = x_1 + x_2 - 8$, $h_2 = -x_1 + x_2 - 5$

$\frac{\partial L}{\partial x_1} = 10 - 2x_1 - \lambda_1 + \lambda_2 = 0$ — (1)

$\frac{\partial L}{\partial x_2} = 10 - 2x_2 - \lambda_1 - \lambda_2 = 0$ — (2)

$\lambda_1 (h_1) = \lambda_1 (x_1 + x_2 - 8) = 0$ — (3)

$\lambda_2 (h_2) = \lambda_2 (-x_1 + x_2 - 5) = 0$ — (4)

$x_1 + x_2 - 8 \leq 0$ — (5)

$-x_1 + x_2 - 5 \leq 0$ — (6)

$x_1, x_2 \geq 0$ — (7)

$\lambda_1, \lambda_2 \geq 0$ max — (8)